A. Division, Remainders, and Congruence

Divisibility

For integers d and n,

$$d \mid n \iff \text{there is a } k \in \mathbb{Z} \text{ such that } n = dk.$$

Here $\mathbb{Z}$ is the set of integers, d is a **divisor** (or factor), n is a **multiple** of d, and k is an integer **witness**. To prove $d \nmid n$, show that no integer witness can exist.

Division Algorithm

For every $a \in \mathbb{Z}$ and positive integer b, there are unique integers q, r such that

$$a = bq + r, \qquad 0 \le r < b.$$

The integer a is the **dividend**, b the **divisor**, q the **quotient**, and r the **remainder**. We write $q = a \operatorname{div} b$ and $r = a \bmod b$. The bound on r makes the answer canonical, even when a is negative.

Congruence

For $m \ge 2$,

$$x \equiv y \pmod{m} \iff m \mid (x - y).$$

The positive integer m is the **modulus**; we may also write $x \equiv_m y$. Equivalently, $x = y + mk$ for some $k \in \mathbb{Z}$, or x and y have the same least nonnegative remainder modulo m.

One relationship, three views

If $a = bq + r$ with $0 \le r < b$, then

$$a \bmod b = r \quad \text{and} \quad a \equiv r \pmod{b}.$$

The equation gives the quotient and remainder; congruence records only the remainder class.

Basic divisibility laws

For integers of the indicated form:

$$\begin{aligned} d \mid a,\ d \mid b &\implies d \mid (ma + nb), \quad m, n \in \mathbb{Z}, \\ d \mid a &\implies d \mid ac, \quad c \in \mathbb{Z}, \\ d \mid a,\ a \mid b &\implies d \mid b. \end{aligned}$$

Each follows by substituting integer witnesses and factoring out d.

Worked example: a positive dividend

Find all three views for 263 divided by 17. **Choose q:** $17(15) = 255$ is the largest multiple not exceeding 263. **Compute r:** $263 - 255 = 8$, so

$$263 = 17(15) + 8, \qquad 0 \le 8 < 17.$$

The bound $0 \le 8 < 17$ makes this canonical. Therefore

$$263 \operatorname{div} 17 = 15, \qquad 263 \bmod 17 = 8, \qquad 263 \equiv 8 \pmod{17}.$$

Congruence check: $263 - 8 = 255 = 17(15)$, so $17 \mid (263 - 8)$.

Worked example: a negative dividend

Find the canonical quotient and remainder for -52 divided by 9. **Reject the first try:** $-52 = 9(-5) - 7$ has a negative remainder. **Adjust:** decrease the quotient by 1 and add 9 to the remainder:

$$-52 = 9(-6) + 2, \qquad 0 \le 2 < 9.$$

Now $0 \le 2 < 9$. Thus

$$-52 \operatorname{div} 9 = -6, \qquad -52 \bmod 9 = 2, \qquad -52 \equiv 2 \pmod{9}.$$

Common mistake

Do not truncate a negative quotient toward zero. Choose the unique q that makes $0 \le r < b$. A negative number may be congruent to a negative residue, but $a \bmod b$ is the least nonnegative one.

Can you do this from the definitions?

Given a and $b > 0$, find q, r; verify the remainder bound; state $a \operatorname{div} b$, $a \bmod b$, and a congruence; justify the congruence by showing that b divides a difference.

Recover a number; test a congruence

Recover: insert $q = -4, r = 9$ into $a = 13q + r$ to get $a = 13(-4) + 9 = -43$; $0 \le 9 < 13$ verifies the remainder. **Test:** subtract in $347 \equiv -13 \pmod{24}$: $347 - (-13) = 360 = 24(15)$, so the congruence is true. Finally, $347 = 24(14) + 11$, so the least nonnegative residue is 11.

B. Base-b Representations

Base-b expansion

For $b \geq 2$, the numeral $(a_k a_{k-1} \cdots a_0)_b$ represents

$$a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0, \qquad 0 \leq a_i < b.$$

Binary uses digits $0, 1$; octal uses 0 through 7; hexadecimal uses 0 through 9 and $A = 10, \ldots, F = 15$.

Place value, shifting, and digits

$$(10)_b = b, \qquad (100)_b = b^2, \qquad (10)_b(10)_b = (100)_b.$$

More generally,

$$\big((10)_b\big)^k = \big(1\underbrace{00\cdots0}_{k \text{ zeros}}\big)_b = b^k.$$

If $N = (a_k \cdots a_0)_b$ and $0 \leq r < b$, then

$$bN = (a_k \cdots a_0 0)_b, \qquad bN + r = (a_k \cdots a_0 r)_b.$$

Equivalently,

$$(a_k \cdots a_1 a_0)_b = b(a_k \cdots a_1)_b + a_0.$$

Thus multiplying by b shifts left, while dividing by b removes the final digit: $\operatorname{div} b$ gives the remaining prefix and $\operatorname{mod} b$ gives the removed digit.

Decimal to another base

Repeatedly divide by the new base. Record each remainder. The digits are the remainders read from *bottom to top*. Verify by expanding in powers of the base. Each line $n = bq + r$ reverses one shift-and-append step. In hexadecimal, remainders 10–15 become A–F; for 1000_{10} the successive remainders are $8, E, 3$, so $1000_{10} = (3E8)_{16}$.

Worked example: 202_{10} to binary

Divide and record remainders:

$$\begin{array}{rcl@{\qquad}rcl} 202 &=& 2(101) + 0 & 101 &=& 2(50) + 1 \\ 50 &=& 2(25) + 0 & 25 &=& 2(12) + 1 \\ 12 &=& 2(6) + 0 & 6 &=& 2(3) + 0 \\ 3 &=& 2(1) + 1 & 1 &=& 2(0) + 1 \end{array}$$

Read upward: the remainders in reverse chronological order give $202_{10} = 11001010_2$. Rebuilding by shifting and appending verifies the digit order:

$$\begin{aligned} (1)_2 &\to (11)_2 \to (110)_2 \\ &\to (1100)_2 \to (11001)_2 \to (110010)_2 \\ &\to (1100101)_2 \to (11001010)_2. \end{aligned}$$

Place-value check: $128 + 64 + 8 + 2 = 202$.

Worked example: evaluate a numeral

Multiply each digit by the matching power of 5:

$$(2301)_5 = 2(5^3) + 3(5^2) + 0(5) + 1 = 326.$$

Thus the value is 326_{10}. The subscript names the base; it is not a multiplier.

Binary as a bridge

One hexadecimal digit equals exactly four binary bits; one octal digit equals exactly three. Pad with leading zeros, then regroup from the *right*. For example,

$$\begin{aligned} (725)_8 &= (111\ 010\ 101)_2 = 469_{10}, \\ (101\ 101\ 011)_2 &= (553)_8, \\ (0001\ 0110\ 1011)_2 &= (16B)_{16}. \end{aligned}$$

Worked example: hexadecimal to binary and octal

Convert $(2D7)_{16}$. **Hex to binary:** replace each digit with four bits:

$$2 = 0010_2, \qquad D = 1101_2, \qquad 7 = 0111_2,$$

so

$$(2D7)_{16} = (0010\ 1101\ 0111)_2.$$

Binary to octal: regroup from the right in threes,

$$001\ 011\ 010\ 111 \implies (1327)_8.$$

Decimal check:

$$2(16^2) + 13(16) + 7 = 727.$$

Leading zeros change grouping, not value.

Worked example: the last two base-b digits

Let $n = (25314)_7$. Then

$$n = 2(7^4) + 5(7^3) + 3(7^2) + 1(7) + 4.$$

Discard higher places: every term containing 7^2 is congruent to 0 modulo 7^2. Therefore

$$n \bmod 7^2 = 1(7) + 4 = 11.$$

Interpret: the last two base-seven digits are $(14)_7$, whose decimal value is $1(7) + 4 = 11$, not 14.

Conversion checks

Digits must be legal for the base. Binary/hex groups have four bits; binary/octal groups have three. Expand the final numeral to verify it represents the same integer.

C–D. Modular Arithmetic, GCD, and Bézout

Arithmetic with congruences

If $a \equiv b \pmod m$ and $c \equiv d \pmod m$, then

$$a + c \equiv b + d \pmod m, \qquad ac \equiv bd \pmod m.$$

In particular, $a^n \equiv b^n \pmod m$ for every positive integer n. The exact remainder identities are

$$(a + c) \bmod m = ((a \bmod m) + (c \bmod m)) \bmod m,$$
$$ac \bmod m = ((a \bmod m)(c \bmod m)) \bmod m.$$

Reduce early; report a least nonnegative residue for $x \bmod m$.

Worked example: reduce early

Compute the least nonnegative residue of $(-47) \cdot 38 + 19$ modulo 12. **Reduce each input:**

$$-47 \equiv 1, \quad 38 \equiv 2, \quad 19 \equiv 7 \pmod{12},$$

Operate on the small representatives:

$$(-47) \cdot 38 + 19 \equiv 1(2) + 7 \equiv 9 \pmod{12}.$$

Since $0 \le 9 < 12$, $((-47) \cdot 38 + 19) \bmod 12 = 9$.

Powers modulo m: cycles and squaring

Cycle method. Compute residues until they repeat. Since $3, 9, 7, 1$ repeat for powers of 3 modulo 10, and $47 \equiv 3 \pmod 4$,

$$3^{47} \equiv 3^3 \equiv 7 \pmod{10}.$$

Repeated squaring. Square and reduce at every step. Modulo 13,

$$5^2 \equiv -1, \qquad 5^4 \equiv 1,$$

so all powers $5^{4j} \equiv 1$. Because $117 = 64 + 32 + 16 + 4 + 1$,

$$5^{117} \equiv 1 \cdot 1 \cdot 1 \cdot 1 \cdot 5 \equiv 5 \pmod{13}.$$

State the cycle length or the exponent decomposition you used.

GCD and relative primality

For integers a, b, not both zero, $\gcd(a, b)$ is the greatest positive integer dividing both. The numbers are **relatively prime** when $\gcd(a, b) = 1$. The numbers themselves need not be prime.

Euclidean algorithm

Put the larger positive input first, then repeatedly use

$$a = bq + r, \qquad 0 \le r < b,$$

then replace (a, b) by (b, r). Stop when the remainder is 0. The last *nonzero* remainder is the gcd. Indeed, from $a = bq + r$, the common divisors of a, b are exactly those of $b, r = a - bq$, so $\gcd(a, b) = \gcd(b, r)$. Swap the inputs first if necessary.

Euclidean algorithm and Bézout coefficients

Find $\gcd(246, 102)$ and express it as $246s + 102t$. **Step 1: run the Euclidean algorithm.**

$$\begin{aligned} 246 &= 2(102) + 42, \\ 102 &= 2(42) + 18, \\ 42 &= 2(18) + 6, \\ 18 &= 3(6) + 0. \end{aligned}$$

The last nonzero remainder is 6, so it is the gcd. **Step 2: back-substitute.** Bézout's identity says that $\gcd(a, b) = sa + tb$ for some $s, t \in \mathbb{Z}$. Back-substitute:

$$\begin{aligned} 6 &= 42 - 2(18) \\ &= 42 - 2\big(102 - 2(42)\big) \\ &= 5(42) - 2(102) \\ &= 5\big(246 - 2(102)\big) - 2(102) \\ &= 5(246) - 12(102). \end{aligned}$$

Thus $s = 5$ and $t = -12$. **Step 3: verify both conclusions:**

$$6 \mid 246, \quad 6 \mid 102, \qquad 5(246) - 12(102) = 6.$$

Bézout and relative-prime checks

Back-substitute through every displayed Euclidean equation; preserve the original input order in the final identity. Coefficients are not unique, so verify them by substitution. For example, $16 = 2^4$ and $45 = 3^2 \cdot 5$ share no prime factor, so $\gcd(16, 45) = 1$: composite numbers can be relatively prime.

E. Definition-Driven Proofs and Quiz Bridges

Even, odd, prime, and composite

An integer is **even** if it is $2k$, and **odd** if it is $2k+1$, for some $k \in \mathbb{Z}$. A positive integer $p > 1$ is **prime** if its only positive divisors are 1 and p; an integer greater than 1 that is not prime is **composite**.

Proof pattern: divisibility and linear combinations

Unpack: assume $d \mid a$ and $d \mid b$, so $a = dr$ and $b = ds$ for some $r, s \in \mathbb{Z}$. **Substitute and factor:** for any $m, n \in \mathbb{Z}$,

$$ma + nb = m(dr) + n(ds) = d(mr + ns).$$

Close: $mr+ns \in \mathbb{Z}$, so the display gives the required witness and $d \mid (ma + nb)$.

Proof pattern: a modulus that factors

Unpack: suppose $p, q \geq 2$, $d = pq$, and $x \equiv y \pmod{d}$. Then $d \mid (x - y)$, so $x - y = kd = kpq$ for some $k \in \mathbb{Z}$. **Refactor:**

$$x - y = p(kq) \quad \text{and} \quad x - y = q(kp).$$

Close: $kq, kp \in \mathbb{Z}$, so $p \mid (x - y)$ and $q \mid (x - y)$, giving

$$x \equiv y \pmod{p}, \qquad x \equiv y \pmod{q}.$$

This direction needs only positive moduli with $p \mid d$ and $q \mid d$; primality is not required. For instance, $103 - 4 = 99$ proves the same claim from modulus 33 to moduli 3 and 11.

A reliable proof skeleton

1. Translate each hypothesis using its definition.
2. Name the integer witnesses.
3. Manipulate the resulting equations.
4. Exhibit the witness required by the conclusion.
5. State why the witness is an integer.

Proof pattern: products preserve congruence

Unpack: assume $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Then $a = b + mk$ and $c = d + m\ell$ for some $k, \ell \in \mathbb{Z}$. **Expand and factor:**

$$ac - bd = m(b\ell + dk + mk\ell).$$

Close: the parenthesized expression is an integer, so $m \mid (ac - bd)$ and $ac \equiv bd \pmod{m}$.

Proof pattern: odd squares modulo 8

Represent the odd integer: let $n = 2k + 1$. **Expand:**

$$n^2 - 1 = 4k(k + 1).$$

Find the extra factor of 2: one of $k, k + 1$ is even, so $k(k + 1) = 2t$ for some $t \in \mathbb{Z}$. Hence $n^2 - 1 = 8t$. Thus $8 \mid (n^2 - 1)$, proving $n^2 \equiv 1 \pmod{8}$.

Two short divisibility arguments

Transitivity: if $a \mid b$ and $b \mid c$, write $b = ar$ and $c = bs$. Substitution gives $c = (ar)s = a(rs)$; since $rs \in \mathbb{Z}$, $a \mid c$.
Three consecutive integers: let $P = n(n + 1)(n + 2)$. One factor is even, so $P = 2r$ for some $r \in \mathbb{Z}$. One factor is divisible by 3, so $P = 3s$ for some $s \in \mathbb{Z}$. Combine these witnesses:

$$P = 3P - 2P = 3(2r) - 2(3s) = 6(r - s).$$

Since $r - s \in \mathbb{Z}$, the definition gives $6 \mid P$.

Claims that need an extra hypothesis

From $a \mid bc$, it does *not* follow in general that $a \mid b$ or $a \mid c$: $6 \mid 2 \cdot 3$, but 6 divides neither factor. The implication is valid when a is prime (Euclid's lemma): if prime $p \mid bc$, then $p \mid b$ or $p \mid c$.

Core readiness and format

A: canonical division and congruence. **B:** base conversion. **C:** modular arithmetic, cycles, squaring, and congruence proofs. **D:** Euclidean algorithm and Bézout. **E:** direct proofs and counterexamples. A live Core designates one problem from each section A–E and lasts 20 minutes. Four fully correct problems earns **Meets**; all five earns **Exceeds**. Every retake is a complete new form. Show enough work for each answer and proof to be checked; approved accommodations apply.